Return Predictions From Trade Flow

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1 Introduction

Here you will assess trade flow as means of generating profit opportunities in BTC or ETH cryptotoken markets at multiple exchanges. We stress the word "opportunity" because at high data rates like these, and given the markets' price-time priority, it is far easier to identify desirable trades in the data stream than it is to inject oneself profitably into the fray.

2 Data

We have high-frequency messages corresponding to $N > 100MM$ trades in BTC-USDT and ETH-USDT, at several exchanges, available on the class website. These are collected into 1GB+ parquet-format files. Loaded into Pandas, this takes 11GB of memory, so we also have a subsample available on the class website with 1/6 the data.

2.1 Treatment

Load the trade data for your choice of ETH or BTC from the class website. For each exchange, split it into test and training sets, with your training set containing the first 40% of the data and the test set containing the remainder.

2.2 Format

The data has the following structure

2.2.1 Trades

	ts	side	qty	trade_price	Exchange
846	2025-05-22 00:00:00.059739309	B	0.000080	109646.000000	OKX
958	2025-05-22 00:00:00.172964441	B	0.000181	109648.000000	OKX
959	2025-05-22 00:00:00.176589875	B	0.000010	109650.000000	OKX
45456560	2025-05-08 23:59:59.457560930	B	0.000080	103261.610000	BINANCE
45456606	2025-05-08 23:59:59.599598458	B	0.000120	103261.610000	BINANCE
45456733	2025-05-08 23:59:59.736459842	A	0.100000	103261.600000	BINANCE

Here, the received time comes from the clock of the recording device, which was not synchronized to the exchange clock. Such inaccuracies in clock settings, i.e. "clock skew", can cause exchange timestamps, when available, to appear later than the time at which they are recorded as having been received.

3 Exercise

Work on one exchange at a time, and as necessary truncate or subsample the data into a small enough data set that your entire notebook can run in 30 minutes or less.

Write code to find τ-interval trade flow $F_i^{(\tau)}$ just prior[1] to each trade data point[2] i. Compute T-second forward returns[3] $r_i^{(T)}$. Regress them against each other in your training set, to find a coefficient β of regression.

For each data point in your test set you already have $F_i^{(\tau)}$, so your return prediction is $\hat{r}_i := \beta \cdot F_i^{(\tau)}$. Define thresholds j for $\hat{r}_i$ and assume you might attempt to trade whenever $j < |\hat{r}_i|$. Good values for j will have relatively frequent participation, but not anywhere near 100%. 5% is a nice target.

4 Analysis

Assess the trading opportunities arising from using these return predictions in your test set, both with and without trading cost assumptions. Make sure to link presumed trade size to printed size.

There are two ways to handle PL for this exercise. One is very easy: assume zero PL beyond T seconds for every trade you participate in . Then total PL is just the T-forward next upcoming trade price minus the original trade price, times quantity , summed over all cases.

Alternatively, you can cumulate all the buys and sells at their respective quantities. At the simulation end, there is a leftover quantity that you can mark to the very last appearing trade price as part of the PL. This lets you analyze tails and drawdowns in a much more useful way.

Examine Sharpe ratios, drawdowns and tails. Check correspondence of profitability with a metric of volatility possessing about a 5 minute time scale. As part of this assessment, comment on the reliability/stability of β (most easily done by further splitting the data set), how you chose j, and what you might expect from using much longer training and test periods.

[1] We do not include the trade i data itself, because we are evaluating trade i in terms of the flow we would have been aware of just before it happened.

[2] NOTE: the trade data series does not necessarily have strictly increasing timestamps. Be sure not to include other trades at the same timestamp in your computation of F_i.

[3] You need not handle latency in your homework, but for your edification: a more careful implementation would account for lags. For a pessimistic approach we could choose L as, say, twice the 99th percentile of computational and communications lag. Then, it would use book data (not just trade data) to help compute return from time $t_i + L$ to $t_i + L + T$ and run regressions using that. The idea here is that it takes approximately time L to "do anything" about trade information.